

ELK Stack Based Security Information and Event Management (SIEM) Dashboard

PART 1 - Front Matter and Introduction

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Certificate

This is to certify that the project entitled **ELK Stack Based Security Information and Event Management (SIEM) Dashboard** submitted by **Gurpreet Singh** is submitted in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications (MCA) under the School of Engineering & Technology, CT University.

Declaration

I, Gurpreet Singh, declare that this project is my original work and has not been submitted previously for any degree or diploma. The work has been completed under academic guidance and all referenced sources have been acknowledged.

Acknowledgement

I sincerely thank the faculty of the School of Engineering & Technology, CT University, for their guidance and encouragement. I also thank my family and friends for their continuous support throughout the project.

Abstract

This project demonstrates the implementation of a Security Information and Event Management (SIEM) platform using the ELK Stack (Elasticsearch, Logstash and Kibana). Docker Compose is used to deploy the complete environment. Logstash ingests and parses sample log files, Elasticsearch indexes the data, and Kibana provides interactive dashboards for monitoring web traffic and security events.

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Chapter 1: Introduction

Cyber Security focuses on protecting information systems, networks, and applications from unauthorized access and cyber threats. Modern organizations generate large volumes of log data from web servers, operating systems, applications, and security devices. A SIEM solution centralizes these logs, enabling efficient monitoring, investigation, and analysis.

What is SIEM?

Security Information and Event Management (SIEM) combines Security Information Management (SIM) and Security Event Management (SEM). It collects, normalizes, stores, and analyzes security logs from multiple sources. SIEM solutions provide centralized visibility, support incident investigation, and help identify suspicious activity.

Project Objectives

- Implement a basic SIEM platform using the ELK Stack.
- Collect and process web server logs.
- Parse logs using Logstash Grok filters.
- Store events in Elasticsearch.
- Create Kibana dashboards for visualization.
- Demonstrate centralized log monitoring using Docker Compose.

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Part 2 – ELK Stack, Literature Review and System Design

Chapter 2: ELK Stack Overview

The ELK Stack consists of Elasticsearch, Logstash and Kibana. Elasticsearch is a distributed search and analytics engine that stores indexed log data. Logstash is a data processing pipeline that collects, parses and forwards logs. Kibana is a visualization platform used to create dashboards and analyze security events. Docker Compose was used to deploy these services together, making installation and management simpler.

2.1 Elasticsearch

Elasticsearch stores structured log data as documents in indices. It provides full-text search, fast queries and aggregations. In this project, processed logs are stored in the 'web-logs' index and queried by Kibana for visualization.

2.2 Logstash

Logstash reads sample web server logs, applies Grok filters to extract fields such as client IP, HTTP method, URL, status code and timestamp, and forwards the structured events to Elasticsearch.

2.3 Kibana

Kibana provides dashboards, Discover, Lens and Visualize tools. The project dashboard displays Requests Over Time, HTTP Status Codes, Top Client IPs, HTTP Methods and Most Requested URLs.

Chapter 3: Literature Review

SIEM platforms are widely used in Security Operations Centers (SOCs) to centralize log collection and incident investigation. The ELK Stack is a popular open-source solution because it is scalable, flexible and supports powerful search and visualization capabilities. Docker enables rapid deployment of ELK services across development environments.

Chapter 4: Hardware and Software Requirements

Hardware Requirements:

- Intel Core i5 or higher
- Minimum 8 GB RAM (16 GB recommended)
- 20 GB free disk space

Software Requirements:

- Windows 11
- Docker Desktop
- Visual Studio Code
- Elasticsearch 8.15
- Logstash 8.15
- Kibana 8.15

Chapter 5: System Architecture

Architecture Workflow:

Sample Logs → Logstash → Elasticsearch → Kibana Dashboard

Logstash ingests and parses logs, Elasticsearch indexes the structured events, and Kibana

visualizes the data using interactive dashboards.

Project Workflow

1. Start Docker Desktop.
2. Run **docker compose up -d**.
3. Logstash reads sample log files.
4. Elasticsearch indexes parsed events.
5. Kibana displays dashboards and Discover views.
6. Analysts monitor activity and investigate logs.

Suggested Diagrams for Final Report

- ELK Stack Architecture Diagram
- Docker Container Diagram
- Deployment Diagram
- Data Flow Diagram (Level 0 & 1)
- Use Case Diagram
- Activity Diagram
- Sequence Diagram
- Component Diagram
- Project Folder Structure Diagram

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Part 3 – Implementation and Configuration

Chapter 6: Docker Installation

Docker Desktop was installed on Windows 11 to run Elasticsearch, Logstash and Kibana inside isolated containers. Docker Compose was used to start all services using a single configuration file.

Chapter 7: Visual Studio Code Setup

Visual Studio Code was used to create and edit the project files including docker-compose.yml, Logstash pipeline configuration and sample log files. The integrated terminal was used to execute Docker commands.

Project Folder Structure

```
project/
├── docker-compose.yml
├── logstash/
│   ├── pipeline/
│   │   └── logstash.conf
│   └── sample_logs/
│       ├── access.log
│       └── auth.log
```

docker-compose.yml

The docker-compose.yml file defines three services: Elasticsearch – stores indexed logs. Logstash – parses and forwards logs. Kibana – visualizes indexed data. A Docker volume is used to persist Elasticsearch data between restarts.

Logstash Pipeline (logstash.conf)

The Logstash pipeline contains three stages: **Input**: Reads log files from the mounted sample_logs directory. **Filter**: Uses Grok patterns to extract fields such as client_ip, request, method, status and bytes. **Output**: Sends parsed events to Elasticsearch index **web-logs** and prints them to the console for debugging.

Elasticsearch Configuration

Elasticsearch runs in single-node mode with security disabled for development. It indexes processed log events and provides REST APIs for searching and aggregating data.

Kibana Configuration

Kibana connects to Elasticsearch using the ELASTICSEARCH_HOSTS environment variable. A data view named **web-logs** was created to explore indexed documents and build visualizations.

Project Execution Commands

Start the ELK Stack

```
docker compose up -d
```

Check running containers

```
docker ps
```

Restart Logstash

```
docker compose restart logstash
```

Stop containers

`docker compose down`

Expected Output

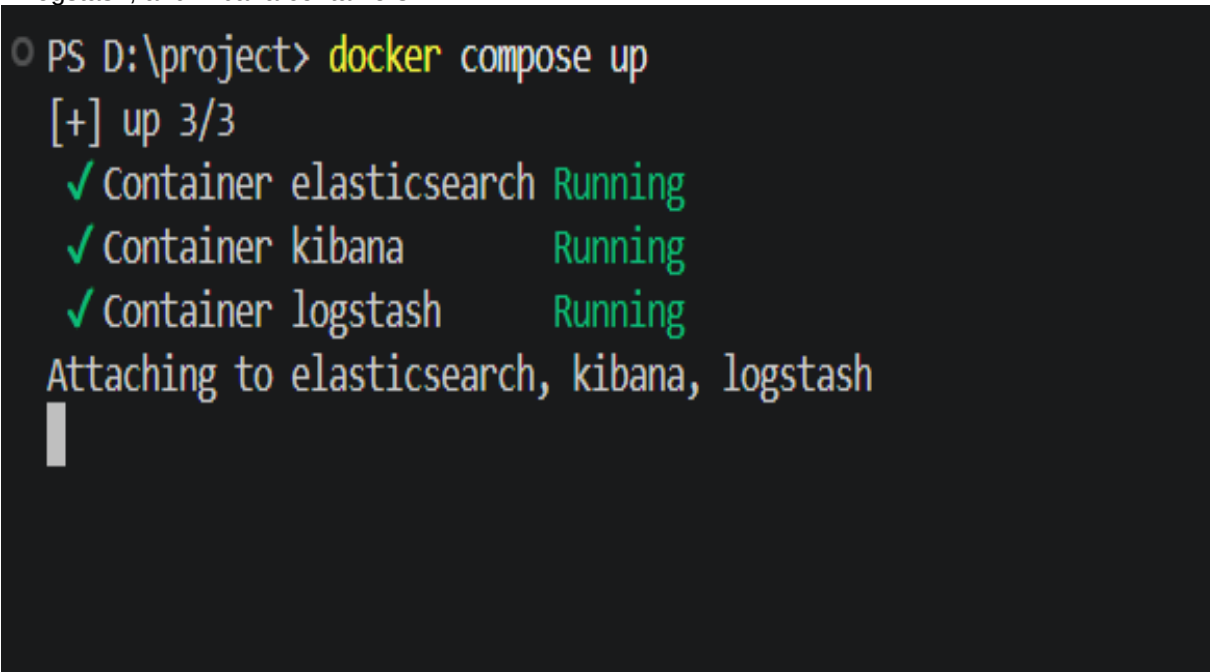
After starting the stack: Elasticsearch is available at <http://localhost:9200> Kibana is available at <http://localhost:5601> Logstash ingests sample logs into the **web-logs** index. Kibana Discover displays indexed documents.

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Part 4 – Project Execution, Screenshots and Dashboard Analysis

Figure 4.1: Starting ELK Stack using Docker Compose.

This screenshot shows the Docker Compose command being executed to start Elasticsearch, Logstash, and Kibana containers.

A terminal window with a dark background. The prompt is 'PS D:\project>'. The command 'docker compose up' is entered. The output shows '[+] up 3/3' followed by three lines: '✓ Container elasticsearch Running', '✓ Container kibana Running', and '✓ Container logstash Running'. Below these, it says 'Attaching to elasticsearch, kibana, logstash' and a cursor is visible on the next line.

```
PS D:\project> docker compose up
[+] up 3/3
✓ Container elasticsearch Running
✓ Container kibana Running
✓ Container logstash Running
Attaching to elasticsearch, kibana, logstash
```

Figure 4.2: Docker Compose Execution.

The services are launched in detached mode. Docker automatically creates the required network and starts each service.

```
PS D:\project> docker compose up -d
[+] up 3/3
✓ Container logstash      Running
✓ Container elasticsearch Running
✓ Container kibana        Running
PS D:\project> |
```

Figure 4.3: Elasticsearch Verification.

Opening <http://localhost:9200> confirms Elasticsearch is running and responding to REST API requests.

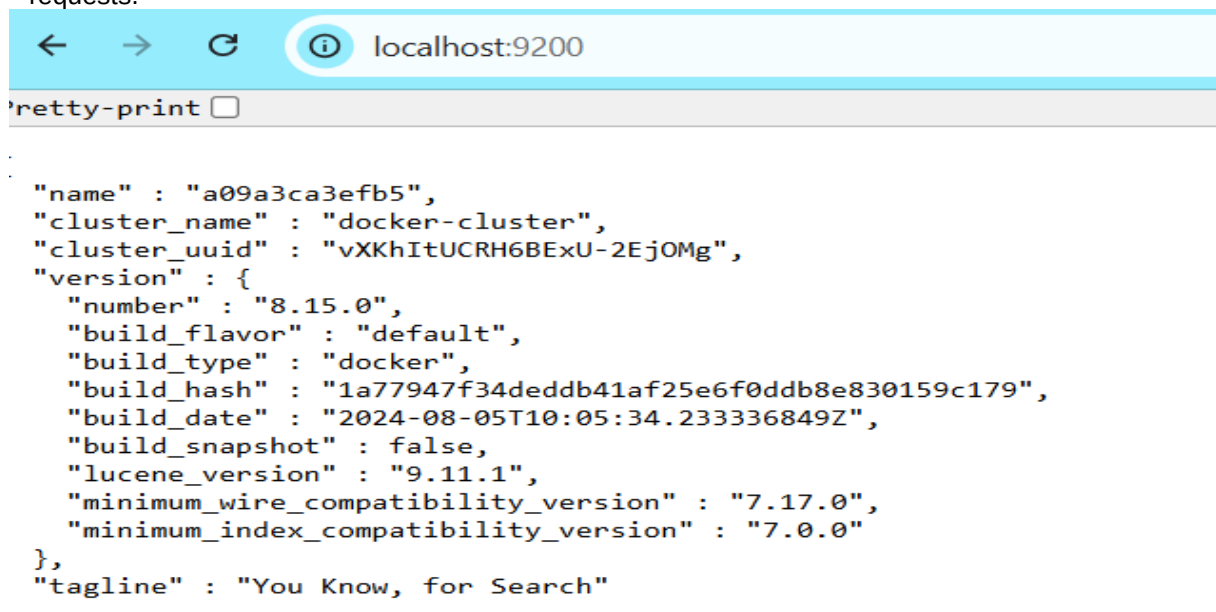


Figure 4.4: Kibana Home Page.

Kibana provides the graphical interface for searching data, creating dashboards, and visualizing events.

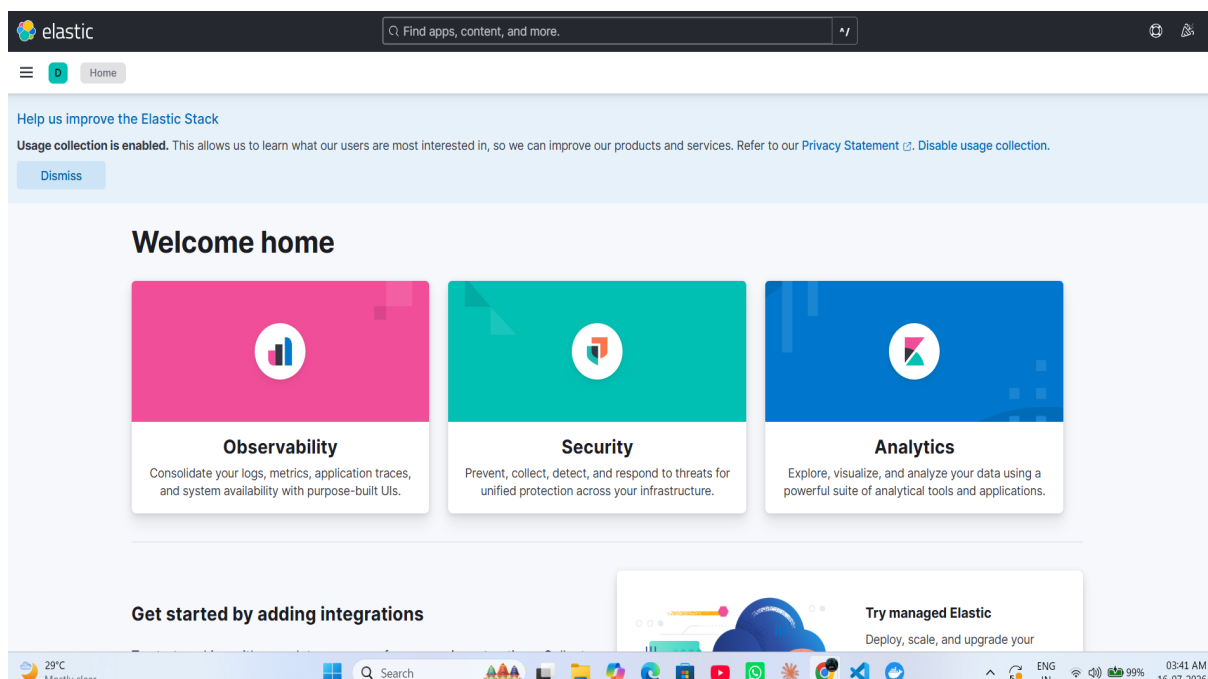


Figure 4.5: Indexed Documents.

The Discover page displays documents stored in the web-logs index after processing by Logstash.

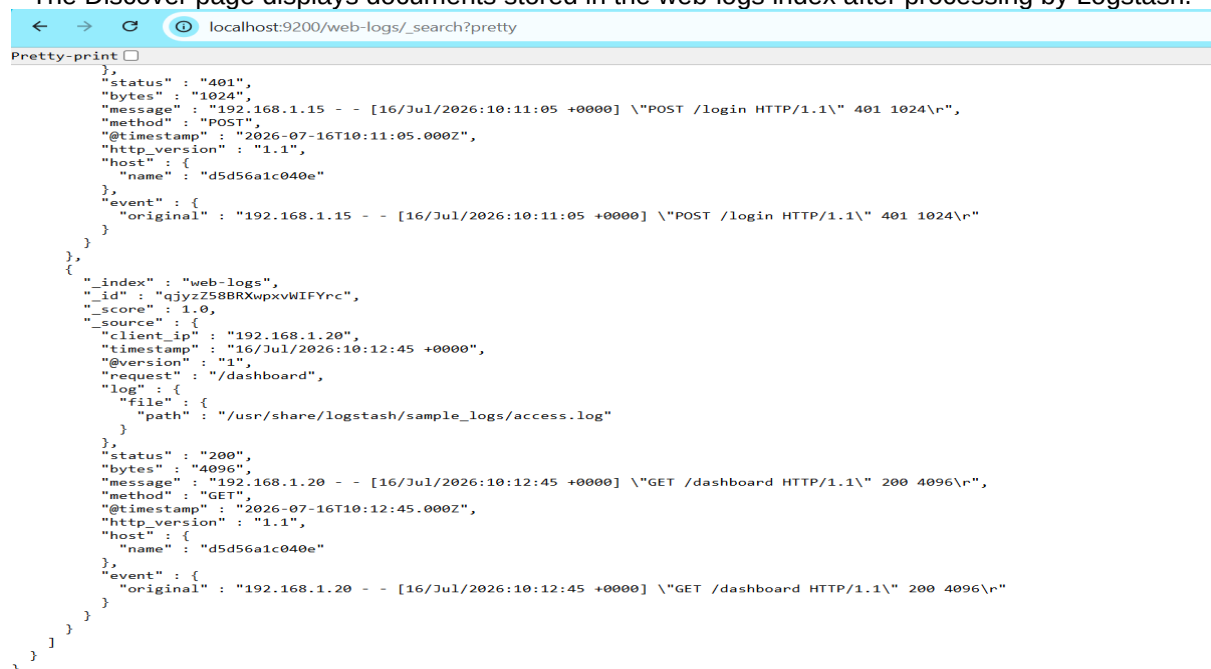


Figure 4.6: SIEM Dashboard.

The dashboard combines multiple visualizations into a single monitoring interface.

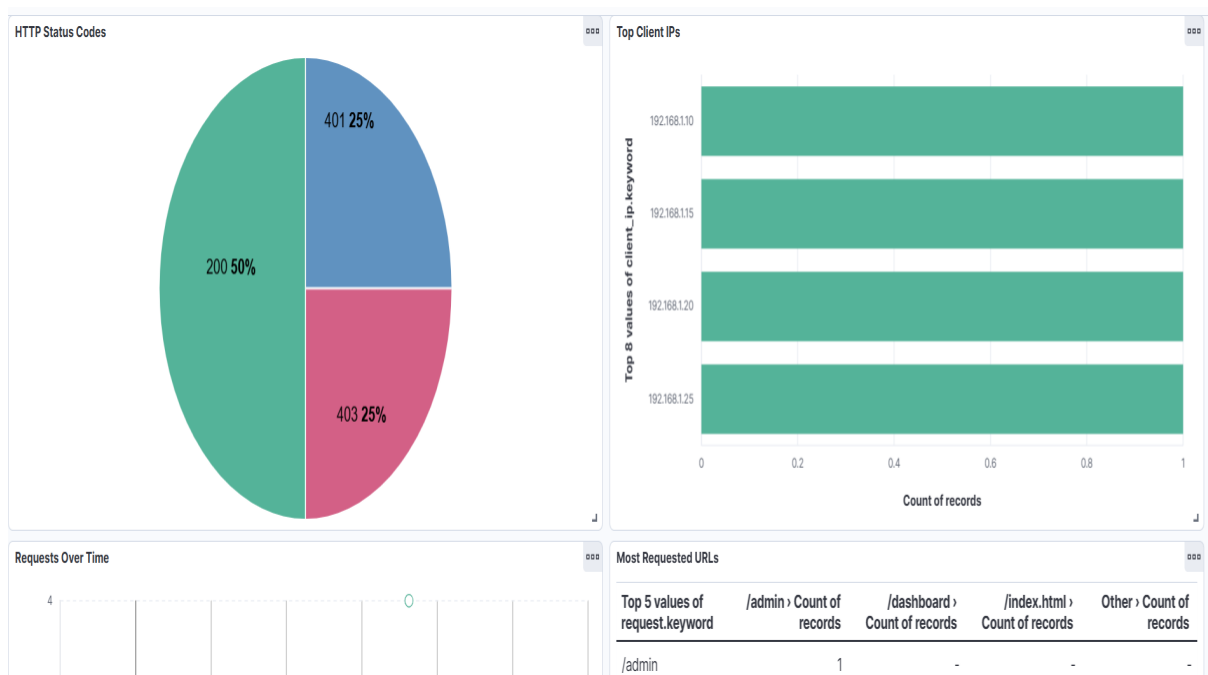


Figure 4.7: HTTP Methods Visualization.

This visualization summarizes GET and POST requests, helping analysts understand request patterns.

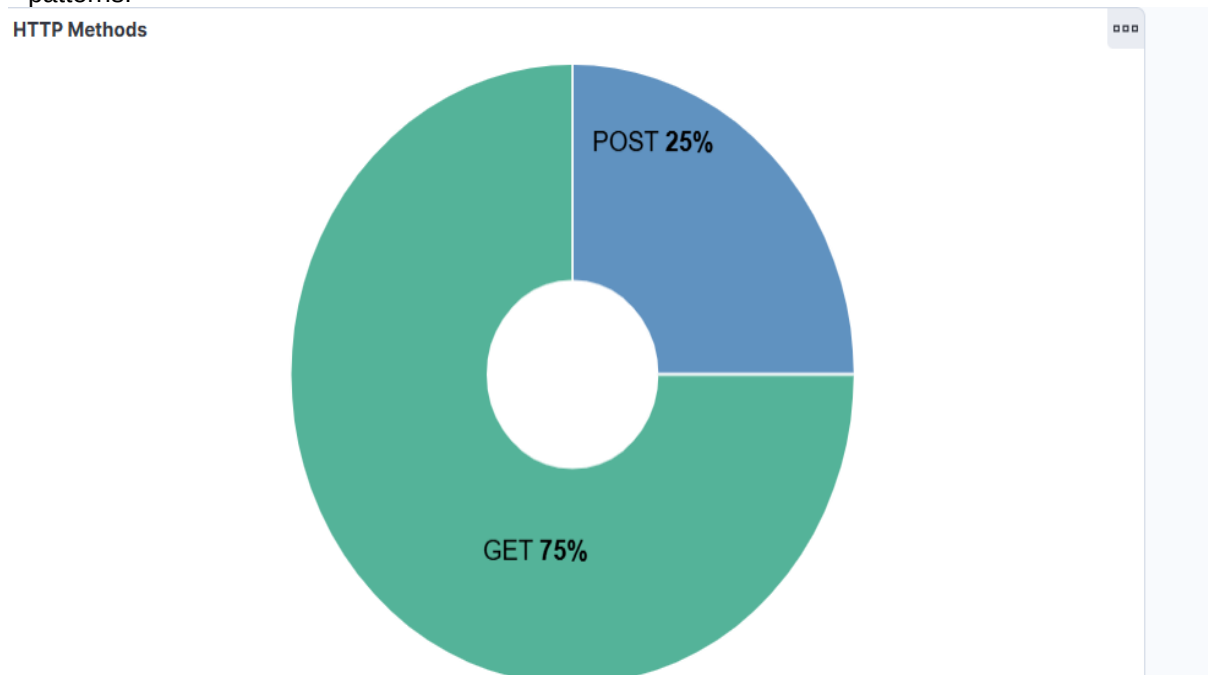


Figure 4.8: Requests Over Time.

A time-series chart displays how requests change over time, useful for identifying spikes in activity.

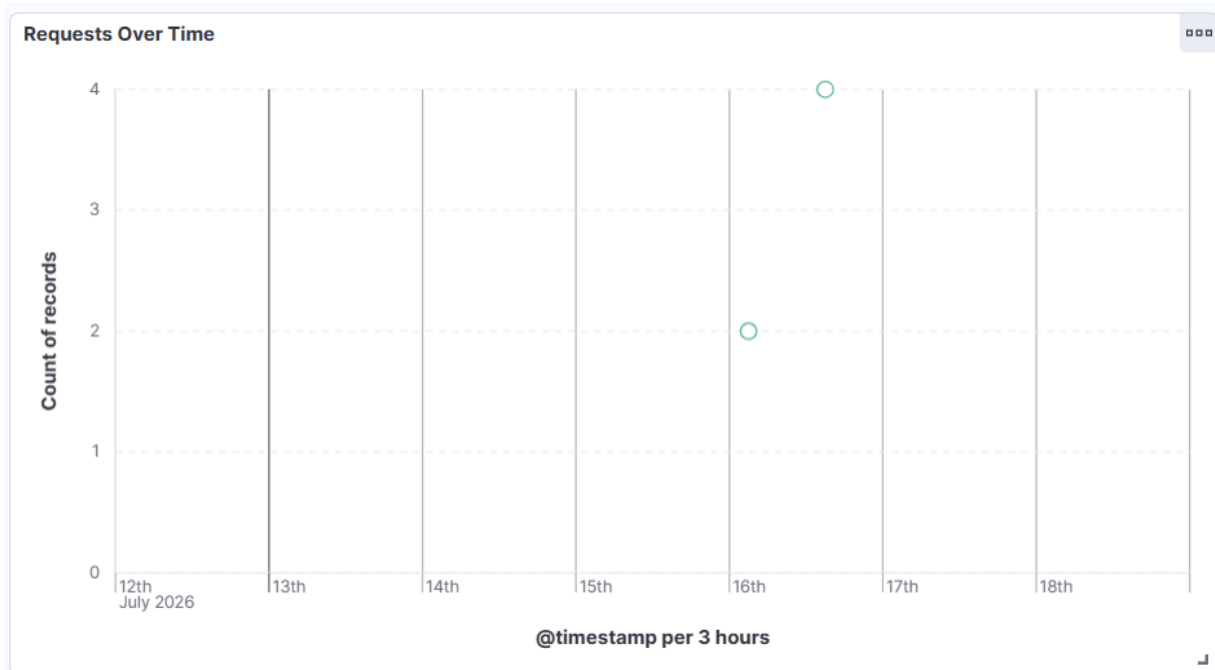


Figure 4.9: Most Requested URLs.

A table lists the most frequently requested resources, helping identify popular or suspicious endpoints.

Most Requested URLs				
Top 5 values of request.keyword	/admin › Count of records	/dashboard › Count of records	/index.html › Count of records	Other › Count of records
/admin	1	-	-	-
/dashboard	-	1	-	-
/index.html	-	-	1	-
/login	-	-	-	1

Dashboard Analysis

The completed SIEM dashboard provides centralized monitoring of web logs. The HTTP Status Code chart highlights successful and failed requests, the Top Client IP chart identifies the most active systems, Requests Over Time visualizes traffic trends, HTTP Methods separates GET and POST activity, and the Most Requested URLs table identifies frequently accessed resources. Together these visualizations demonstrate how the ELK Stack can be used to collect, process, index, and analyze security-relevant log data in a single interface.

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Part 5 – Testing, Results, Conclusion and References

Chapter 8: Testing and Results

The project was tested after deploying Elasticsearch, Logstash and Kibana using Docker Compose. Sample web server logs were successfully parsed by Logstash and indexed into Elasticsearch. Kibana Discover displayed the indexed documents and the custom dashboard visualized the collected data. The tests confirmed that the ELK stack was functioning correctly and that logs could be searched and analyzed efficiently.

Functional Test Cases

Test 1: Start ELK Stack using `docker compose up -d` – **Result:** Passed

Test 2: Verify running containers using `docker ps` – **Result:** Passed

Test 3: Access Elasticsearch on `http://localhost:9200` – **Result:** Passed

Test 4: Access Kibana on `http://localhost:5601` – **Result:** Passed

Test 5: Verify indexed logs in Discover – **Result:** Passed

Test 6: Verify dashboard visualizations – **Result:** Passed

Advantages

- Centralized log collection and monitoring.
- Fast indexing and search using Elasticsearch.
- Interactive dashboards in Kibana.
- Open-source and cost-effective solution.
- Easy deployment with Docker Compose.
- Suitable for cybersecurity learning and demonstrations.

Limitations

- Uses sample log data instead of production data.
- Basic Logstash parsing rules only.
- No automated alerting configured.
- No machine learning or threat intelligence integration.
- Designed primarily for educational purposes.

Future Scope

Future improvements include integrating Filebeat, Winlogbeat, Suricata IDS, firewall logs, email alerting, Elastic Security, anomaly detection, machine learning, and threat intelligence feeds to build a more advanced enterprise SIEM solution.

Conclusion

The ELK Stack Based SIEM Dashboard successfully demonstrates how security logs can be collected, parsed, indexed and visualized using open-source technologies. Docker Compose simplified deployment, Logstash processed the log files, Elasticsearch stored and searched the events, and Kibana provided meaningful dashboards. The project meets its objectives and provides

practical knowledge of SIEM implementation for MCA Cyber Security students.

References

1. Elasticsearch Official Documentation.
2. Logstash Official Documentation.
3. Kibana Official Documentation.
4. Docker Official Documentation.
5. Elastic Stack Documentation.
6. OWASP Logging Cheat Sheet.
7. NIST Cybersecurity Framework.

Appendix – Useful Commands

Start Project

`docker compose up -d`

Stop Project

`docker compose down`

Restart Logstash

`docker compose restart logstash`

View Running Containers

`docker ps`

View Logstash Logs

`docker logs logstash`

Open Elasticsearch

`http://localhost:9200`

Open Kibana

`http://localhost:5601`